# Reddit users unleashed - understanding user behaviour and their impact on meme stocks

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**Abstract**

Online social networks play an important role in financial markets, as was shown vividly in the recent Gamestop hype on Reddit. In the literature either a theoretical model is build to characterize the role of individual social network users on financial markets or aggregated empirical models are tested on small networks, leaving a gap for investigating this relationship empirically for large scale networks. Our study analyzes WallStreetBets using the proposed Interpretable Social Network and Market Behaviour Model (ISBM), which links individual user behavior to market movements. We find that 1) both influential users as well as followers impact market behaviour, 2) changes in the network are observed well before the market frenzy, 3) changes in users behaviour can explain the market movements and improves the adj. R2 from modelling the return by 630%. Our results lead to insights and recommendations for regulators on effects of meme stocks on market

**Keywords:** Social Network Market Impact, Social Network Modelling, Network dynamics, Wallstreetbets, Changepoint detection, meme stocks

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